AHU-200 Air Handling Unit - Maintenance Manual

1. About this manual

Synthetic reference text written for the AlixAI facility-maintenance reference application.
No real equipment is described.
The AHU-200 is a single-duct air handling unit with a supply fan, a return filter bank, a heating coil and a cooling coil.

2. Safety

- Isolate and lock out the unit at the local disconnect before opening any access panel.
- Wait 5 minutes after isolation for the fan motor capacitor to discharge.
- Wear gloves when handling filters: they hold dust and mould spores.

3. Routine maintenance schedule

- Weekly: check the filter differential pressure display and the condensate drain for blockage.
- Monthly: check the supply fan belt tension and look for cracks or glazing on the belt.
- Quarterly: replace the pleated return filter (part FLT-204) every 3 months, or earlier when the differential pressure exceeds 250 Pa.
- Annually: clean both coils, check the damper linkages and test the freeze protection thermostat.

4. Fault codes

- E01 Supply fan overload: check the belt tension, the motor current and the bearings.
- E04 Filter alarm: the differential pressure across the filter bank is above 250 Pa. Replace the filters (FLT-204).
- E07 Coil freeze protection: the coil air temperature fell below 3 C. Check the heating valve and the outside air damper.
- E09 Return air damper fault: the damper did not reach its position within 90 seconds. Check the actuator and the linkage.

5. Replacing the supply fan belt

Use a V-belt B-section 1180 mm (part BLT-11). Loosen the motor base bolts, remove the old belt, fit the new belt and tension it until a firm thumb press at the middle of the span deflects it by 10 mm. Tighten the bolts and run the fan for 10 minutes, then re-check the tension.
A new belt stretches during its first hours of running: re-check the tension after 24 hours of operation.

6. Parts

- FLT-204 Pleated air filter 592x592, quantity per change: 2.
- BLT-11 V-belt B-section 1180 mm, quantity per change: 1.
- CAP-45 Run capacitor 45 uF for the supply fan motor.